For any observable A.

$$\sigma_A^2 = \langle (\hat{A} - \langle A \rangle)\Psi \,|\, (\hat{A} - \langle A \rangle)\Psi \rangle = \langle f | f \rangle$$

where $f = (\hat{A} - \langle A \rangle)\Psi$

for any other observable B.

$$\sigma_B^2 = \langle g | g \rangle \quad \text{where} \quad g = (\hat{B} - \langle B \rangle)\Psi.$$

the Schwarz inequation

$$\sigma_A^2 \sigma_B^2 = \langle f | f \rangle \langle g | g \rangle \geq |\langle f | g \rangle|^2$$

for any complex number z:

$$|z|^2 = [\mathrm{Re}(z)]^2 + [\mathrm{Im}(z)]^2 \geq [\mathrm{Im}(z)]^2 = \left[\frac{1}{2i}(z - z^*)\right]^2$$

$$\therefore \quad \sigma_A^2 \sigma_B^2 \geq \left(\frac{1}{2i}[\langle f | g \rangle - \langle g | f \rangle]\right)^2$$

$$\langle f | g \rangle = \langle (\hat{A} - \langle A \rangle)\Psi \,|\, (\hat{B} - \langle B \rangle)\Psi \rangle = \langle \Psi | (\hat{A} - \langle A \rangle)(\hat{B} - \langle B \rangle)\Psi \rangle$$

$$= \langle \Psi | \hat{A}\hat{B} - \hat{A}\langle B \rangle - \hat{B}\langle A \rangle + \langle A \rangle \langle B \rangle \Psi \rangle$$

$$= \underbrace{\langle \Psi | \hat{A}\hat{B}\Psi \rangle}_{\langle \hat{A}\hat{B} \rangle} - \langle B \rangle \underbrace{\langle \Psi | \hat{A}\Psi \rangle}_{\langle A \rangle} - \langle A \rangle \underbrace{\langle \Psi | \hat{B}\Psi \rangle}_{\langle B \rangle} + \langle A \rangle \langle B \rangle \underbrace{\langle \Psi | \Psi \rangle}_{1}$$

$$= \langle \hat{A}\hat{B} \rangle - \langle A \rangle \langle B \rangle.$$

similarly, $\langle g | f \rangle = \langle \hat{B}\hat{A} \rangle - \langle B \rangle \langle A \rangle$.

$$[\hat{A}, \hat{B}] \equiv \hat{A}\hat{B} - \hat{B}\hat{A}$$

$$\therefore \quad \langle f | g \rangle - \langle g | f \rangle = \langle \hat{A}\hat{B} \rangle - \langle \hat{B}\hat{A} \rangle = \langle [\hat{A}, \hat{B}] \rangle.$$

uncertainty principle :

$$\sigma_A^2 \sigma_B^2 \geq \left(\frac{1}{2i}\langle [\hat{A}, \hat{B}] \rangle\right)^2$$